MIL-R-23764B
8 November 1968
SUPERSEDING
MIL-R-23764A(MC)
10 October 1967

MILITARY SPECIFICATION

RADIO SET: AN/PRC-47

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements for one type of two-man team pack, high frequency, single sideband 100-watt (w) peak envelope power (PEP) transceiver consisting of a receiver-transmitter, an antenna system, transit case and accessories.

2. APPLICABLE DOCUMENTS

*2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification for.
MIL-P-15024	- Plate, Identification-Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-E-16400	- Electronic Equipment, Naval Ship and Shore, General Specification.
MIL-E-17555	- Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-R-22732	- Reliability Requirements for Shipboard and Ground Electronic Equipment.

U. S. ARMY NATICK LABORATORIES

IP/DES S-42-7 Frame, Rucksack, Steel.

FSC 5820

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STANDARDS

MILITARY

MIL-STD-108	- Definition of and Basic Requirement for Enclosure for Electric and Electronic Equipment.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
*MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
*MIL-STD-463	- Definition and System of Units, Electromagnetic Interference Technology.

DRAWINGS

MARINE CORPS

DL 5116-522-2433-011 - Radio Set: AN/PRC-47

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

*3.1 First article. Unless otherwise specified (see 6.2), prior to submission of any complete production items for acceptance, the contractor shall furnish two first articles for determination of conformance to this specification. Examination and tests shall be as specified herein. Approval of the first articles by the procuring activity shall not relieve the contractor of the obligation to supply radio sets conforming to this specification. Any changes or deviations of the production items from the first articles shall be subject to approval of the contracting officer. The first articles will not be included in the quantity to be delivered to the Government.

3.1.1 Function of equipment. The AN/PRC-47 shall have the capability for single sideband voice, continuous wave (CW), and frequency shift keying (FSK) ground radio communications in the frequency range of 2.000 to 11.999 MegaHertz (MHz) and shall be operationally compatible with the radio set, AN/TRC-75. The equipment shall provide for direct reading frequency

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control with a meter indication to aid tuning and antenna loading. Ten thousand (10,000) channels spaced 1 kiloHertz (kHz) apart shall be provided. The equipment shall break down into two packages, each package shall be capable of being quickly attached to a rucksack frame. One package shall contain the receiver-transmitter and the other shall contain the battery power supply, antenna system and accessories for portable operation. A watertight transit case shall be provided and be capable of housing all components and accessories.

3.1.2 Construction. The transceiver unit shall be enclosed in a watertight case with an easily removable control panel cover. When the control panel cover is removed, it shall expose a splashproof control panel. All basic components shall be splashproof when assembled in operating condition or constructed of materials which, when exposed to the elements, shall not adversely affect the operating capability. Convection cooling of the equipment shall be used throughout. The connection to external power sources shall be through a watertight military standard type receptacle. An insulated fitting shall be provided on the transceiver unit for connection to, and support of, a whip antenna. Adjustable and detachable legs shall be provided to support the transceiver during normal field operation. With the transceiver case removed, it shall be possible to remove all modules except the power amplifier from the chassis. Sufficient test points shall be provided on module tops for efficient trouble shooting and alignment work as indicated herein. Modules shall be provided with retaining screws and Cannon type series D connectors, or equal on the bottom side to mate with chassis connectors when the module is mounted in place.

3.1.2.1 Components. The receiver-transmitter, R-671/PRC-47, shall consist of the components specified herein. Each module shall have sufficient test points available to facilitate trouble shooting. Printed circuit boards shall have cabled connecting wires so that they may be easily removed from the module container, and the set may be placed in operating condition.

3.1.2.1.1 Chassis, CH-474/PRC-47. The main chassis CH-474/PRC-47 shall provide support for the various controls and mechanical linkages located on the front panel, intermodule wiring, miscellaneous components, and a nominal 400 hertz (Hz) power oscillator for converting a direct current (dc) input to a 26 volts (v) peak square wave. The power amplifier section shall contain the power amplifier tube with controls and circuitry to match a 15-foot whip antenna or its equivalent to the power tube.

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3.1.2.1.2 Radio frequency (RF) oscillator, O-1032/PRC-47. The RF oscillator module O-1032/PRC-47 shall contain the 3 MHz frequency standard and additional locked oscillators to provide the 100 and 500 kHz outputs.

3.1.2.1.3 Oscillator control, C-4311/PRC-47. The oscillator control module C-4311/PRC-47 shall contain two 10-crystal oscillators and associated mixer, a harmonic selector and associated mixer, and a discriminator. This circuitry is used to stabilize the variable frequency oscillator (VFO) in the data translator module.

3.1.2.1.4 Signal data translator, CV-1377A/PRC-47. The signal data translator module CV-1377A/PRC-47 shall convert the intermediate frequency (IF) to channel frequency and supply a signal of sufficient level to drive the power amplifier. In receive condition the incoming signal shall be converted to the IF.

3.1.2.1.5 Amplifier modulator, AM-3507/PRC-47. The amplifier modulator module AM-3507/PRC-47 shall contain the balanced modulator; automatic load control (ALC) amplifier, IF amplifiers, product detector, and transmit-receive (TR) relay.

3.1.2.1.6 Audio amplifier, AM-3506/PRC-47. The audio amplifier module AM-3506/PRC-47 shall contain the microphone amplifier, speech processing circuitry, audio output amplifier, voice operated circuitry, automatic gain control (AGC) circuitry, and the 800 cycle oscillator for continuous wave (CW) operation.

3.1.2.1.7 Power supply, PP-3518/PRC-47. The power supply module PP-3518/PRC-47 shall supply all alternating current (ac) or dc voltages needed to operate the equipment from either a 400-Hz 115-v external source or the 26-v square wave at 400 Hz from the power oscillator on the main chassis.

3.1.2.1.8 Case, CY-3762/PRC-47. The receiver-transmitter case CY-3762/PRC-47 shall be of aluminum and balsa sandwich construction. When the front panel cover, CW-647/PRC-47, is in place, the receiver-transmitter shall be waterproof.

3.2 Cases. The transit case and transceiver case shall be watertight. With the transceiver front panel cover removed, the radio set shall be splashproof as specified herein.

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3.2.1 Treatment and painting. Treatment and painting of the cases shall conform to type A of MIL-T-704. Color shall match Marine Corps green color No. 23, semigloss (see 6.3).

*3.2.2 Weight. The transceiver as packaged for man-pack, including the rucksack frame, shall not exceed 46.5 pounds.

3.2.3 Environmental conditions.

3.2.3.1 Service conditions. The equipment shall be capable of performing as specified under the service conditions imposed by field employment. These conditions include:

(a)	Ambient temperature range	-40° Centigrade (C) to +60° C
(b)	Ambient humidity range	0 to 100 percent relative
(c)	Vibration	Para 4.2.2.1 applies
(d)	Shock	Para 4.2.2.2 applies
(e)	Water splash	Para 4.3 applies
(f)	Water immersion	Para 4.4 applies
(g)	High temperature	Para 4.5 applies
(h)	Low temperature	Para 4.6 applies
(i)	Temperature shock	Para 4.7 applies
(j)	Humidity	Para 4.8 applies
(k)	Altitude	Para 4.9 applies
(l)	Salt spray	Para 4.10 applies
(m)	Sand and dust	Para 4.11 applies

3.2.3.2 Storage conditions. The equipment shall be capable of being stored under the following conditions (less battery) with a subsequent period of operation:

(a)	Ambient temperature range	-62° C to +65° C
(b)	Altitude	Sea level to 30,000 feet

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3.2.3.3 Standard conditions. Standard conditions are defined as:

(a) Room anbient temperature, 20° C to 40° C.

(b) No shock or vibration present.

(c) Local altitude and relative humidity.

(d) Supply voltage of 24 v.

3.2.4 Stability.

3.2.4.1 Electrical. The equipment, except for the battery power supply, shall operate as specified herein for at least 200 hours, either continuously or intermittently, independent of component failure without necessity for readjustment of any controls other than those located on the front panel.

3.2.4.2 Mechanical. All controls and adjustments shall be designed so their position of adjustment will not be altered when the equipment is subjected to the specified service conditions.

3.2.4.3 Frequency. The frequency stability of the complete equipment after warm-up for 20 minutes under any of the service conditions within the specified range shall be such as to maintain operating frequencies within ±25 Hz over a period of 60 days duration except that at -40° C to -30° C the stability may degrade to ±40 Hz. At factory ambient conditions, the stability shall be within ±15 Hz over a period of 60 days. It shall be possible by means of factory adjustment to re-establish this stability.

3.2.5 Total operating life. The equipment shall have a minimum total operating life of 3,000 hours with reasonable servicing and replacement of parts.

3.2.5.1 Reliability. Reliability shall receive major emphasis during the fabrication and construction of the radio set. This emphasis shall be reflected in the assembly process. The equipment shall be capable of demonstrating a mean time between failure of 230 hours. MIL-R-22732 shall be used in establishing the reliability requirements.

3.2.6 Interchangeability. All components and parts of the equipment shall be physically, functionally, mechanically, and electrically interchangeable with the corresponding components and parts of the Government furnished AN/PRC-47.

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3.2.7 Nameplates. Nameplates and nomenclature shall be in accordance with MIL-P-15024.

3.2.7.1 Operator instruction plate. The operator instruction plate shall set forth the sequence of operation to place the equipment in operation. A coarse indication of the position of tuning and loading controls versus frequency for operation with the whip antenna and the various length of the wire antenna shall be provided.

3.2.8 Identification marking. Modules and module subassemblies shall be identified with the following information rubber stamped or silk screened on a fixed surface, if available, or on a protected tag affixed in a manner so as not to impair operation of the assembly or be easily torn off:

(a) Nomenclature.
(b) Name, if applicable.
(c) Additional information such as Federal stock number, Drawing No. etc., if space permits.

3.2.9 Moisture and fungus treatment. The equipment shall be fungus proofed by selection of parts and materials as nonnutrient for fungus or the parts and materials shall be treated prior to their use in the equipment so that overall spraying of the equipment is not necessary.

3.2.10 Input power requirements. The equipment shall perform as specified when operated from any power source with the following nominal characteristics, with no internal changes required in the transceiver:

(a) A 16-cell silver-zinc battery, BB-451/U.

(b) A nominal 27.5-v dc vehicle system within the limits of 22 v to 28 v.

(c) A nominal 115-v, 400 Hz, single-phase power source within the limits of:

(1)	Voltage	105-125 v
(2)	Frequency	380-420 Hz

3.2.10.1 Power consumption. The power consumption with standard condition input voltage 24 v dc, shall be not greater than the amount listed below for the conditions indicated using a transmitter dummy load of 50 ohms:

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(a) Transmit (voice, low power) 150 w
(b) Transmit (voice, high power) 300 w
(c) Transmit (two tones, low power) 160 w
(d) Transmit (two tones, high power) 340 w
(e) Receive 15 w

3.2.11 Modes of operation. The system shall be capable of transmission and reception in upper sideband suppressed carrier, voice, CW telegraphy and FSK.

3.2.11.1 Upper sideband. Upper sideband shall mean transmission and reception of energy in the frequency spectrum extending from 300 to 3,000 Hz above indicated frequency. The lowest audio frequency in a transmission shall be translated into the lowest RF frequency transmitted.

3.2.11.2 CW telegraphy. CW telegraphy shall mean transmission and reception by means of a single tone keyed on-off on a frequency of 800 Hz above the indicated frequency by use of a hand key.

3.2.11.3 FSK operation. During the FSK operation, audio tones will be supplied to the radio set from an external frequency shift converter through the adapter, U-239/PRC-47.

3.2.12 Antenna tuning and loading. A meter shall be provided on the transceiver front panel to indicate power output. This meter shall provide information for adjusting the power amplifier tuning and loading knobs for maximum power into the antenna. The receiver shall be capable of being tuned without breaking radio silence. The antenna shall be tuned and loaded at reduced power.

3.2.13 Monitoring sidetone. Bias voltage from the RF power amplifier shall close a diode gate passing the output of the microphone amplifier back through the audio amplifier for use as a monitoring sidetone.

3.2.14 Voice operated transmission. After operation of the push-to-talk switch, an audio input signal shall be necessary before full power is applied to the transmitter.

3.2.15 Remote control. The transceiver shall be capable of being controlled, push-to-talk and on-off, over 2 miles of field wire, WD-1/TT, using standard available remote control units such as control group, AN/GRA-6.

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3.2.16 Terminations.

3.2.16.1 Primary power. The primary power termination on the transceiver shall be an MS type watertight multi-pin receptacle with such internal connections as to allow connection to either a 26-v dc power source or a 400-Hz ac source without internal or external adjustment.

3.2.16.2 Audio terminations. The audio terminations on the transceiver shall be two standard 10-pin audio connectors U-79/U.

3.2.16.3 Telegraph terminations. The telegraph and FSK termination on the transceiver shall be in the same connector as the audio termination.

3.2.16.4 Antenna connections. An insulated fitting for connection to and support of a whip antenna and connection to a long wire antenna shall be provided.

*3.2.17 Interference. The equipment shall meet the design requirements and interference control limits of class 1 of MIL-STD-461 with the following exceptions and those listed in Table I:

(a) On conducted and radiated interference tests the fundamental transmitted frequency and harmonics thereof shall not be measured.

(b) On antenna conducted interference in the receive condition the maximum output shall not exceed 200 microvolts (uv). In the transmit condition, the equipment specification shall govern.

(c) Delete intermodulation tests.

(d) On conducted susceptibility the channel frequency and image frequency will not be measured. The audio frequency will not be measured. The audio frequency variation on the power input line shall be not less than 1 v and the spurious tone output shall be not greater than -30 decibel (db) from the reference tone at rated power output.

*(e) On radiated susceptibility tests the channel frequency shall not be measured. Terms and definitions for interference requirements shall be defined in accordance with MIL-STD-463.

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Table I - Interference Exceptions

Interference test type	Channel Freq MHz	Freq MHz & type interference	Spec deviation granted to no more than
Conducted	2.225	.15 broadband	119 db/MHz
Conducted	11.775	.15 broadband	119 db/MHz
Conducted	11.775	12.25 CW	39 db/uv
Conducted	11.775	14.0 CW	40 db/uv
Radiated	2.225	.15 broadband	84 db/MHz
Radiated	2.225	.3 broadband	75.5 db/MHz
Radiated	2.225	.45 broadband	75 db/MHz
Radiated	11.775	.15 broadband	84 db/MHz
Radiated	11.775	.3 broadband	77 db/MHz
Radiated	11.775	.45 broadband	75 db/MHz
Radiated	11.775	12.2724 CW	24.8 db/uv
Radiated	11.775	12.7728 CW	20.5 db/uv
Receiver, front end rejection	Channel freq	Interference freq	Spec deviation granted to no less than
	11.775 MHz	11.7796 MHz	55 db

3.3 Performance. The performance of the equipment shall be as set forth below. Before commencing any performance tests the equipment shall be operated for a period of at least 20 minutes during which time each control shall be operated at least three times.

3.3.1 Frequency channels. By means of the frequency selection controls it shall be possible to select any frequency in the range 2.000 to 11.999 MHz in 1 kHz increments. These frequencies shall correspond with an arithmetic series 2.000; 2.001; 2.002 to 11.999 MHz. Transmission and reception shall be performed on the same frequency, but not simultaneously.

3.3.2 Receiver characteristics.

3.3.2.1 Sensitivity. The sensitivity of the receiver shall be such that with an RF input of 2.0 uv applied across the receiver terminals with frequency adjusted to give a single tone audio output signal of 1,000 Hz, the audio output shall

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be 50 milliwatts minimum and the signal-plus-noise to noise ratio shall be at least 10 db. This requirement will not apply to the upper three channels in each MHz band, i.e., 2.997, 2.998, 2.999, 3.997, 3.998, 3.999 MHz, etc.

3.3.2.2 Selectivity. The overall selectivity of the receiver shall be such that the response to signals from 300 to 3,000 Hz above the selected channel frequency shall be not more than ±6 db from the response to a signal 1,700 Hz above the channel frequency. The -60 db bandwidth shall not be greater than 6 kHz.

3.3.2.3 Response to spurious signals.

3.3.2.3.1 IF rejection. The receiver shall present an attenuation of not less than 80 db relative to the response of the desired signal to a signal on the IF of the receiver.

3.3.2.3.2 Image rejection. When receiving on any channel the receiver shall attenuate images and other spurious signals as listed in the ranges below:

(a) Not less than 80 db relative to the response of a 5 uv desired signal over the range of 2.000 to 5.999 MHz.

(b) Not less than 60 db over the range of 6.000 to 9.999 MHz.

(c) Not less than 50 db over the range of 10.000 to 11.999 MHz.

3.3.2.3.3 Cross modulation. The audio level produced by an undesired RF signal of 1 v, 50 percent modulated with 400 Hz, at the antenna shall be at least 10 db below the audio output of a desired RF signal of not more than 100 uv when the undesired signal is separated from the desired frequency by not more than 10 percent of the desired frequency.

3.3.2.3.4 Other. All other spurious responses shall be at least -60 db relative to the desired signal.

3.3.2.4 AGC characteristics. The AGC action shall be such that when the signal input is varied from 5 to 100,000 uv, the audio output power shall not increase by more than 10 db.

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3.3.2.5 Audio characteristics.

3.3.2.5.1 Audio output. The receiver shall be capable of producing an audio output of 500 milliwatts (MW). With minimum specified RF input, the audio output shall be at least 50 MW.

3.3.2.5.2 Audio distortion. When an RF input signal having a level of 1,000 uv is applied, the total harmonic distortion at 500 MW output shall not exceed 15 percent.

3.3.2.5.3 Overall audio response. The audio response over the range of 300 to 3,000 Hz shall not vary more than ±6 db, relative to the response at 1,700 Hz.

3.3.2.6 Volume control. A volume control shall be provided on the front panel which shall be capable of producing a minimum change in receiver audio output of 45 db.

3.3.2.7 Oscillator radiation. No signal exceeding 800 microwatts shall be emitted from the antenna when the radio is in the receive condition. This may be measured as input into a 50 ohm device.

3.3.2.8 Desensitization. Receiver desensitization due to the presence of strong undesired signals shall be kept at a minimum by employing the maximum selectivity in the RF circuits consistent with other circuit requirements.

3.3.3 Transmitter characteristics.

3.3.3.1 Output. The transmitter, under service conditions shall be capable of producing power as specified in 3.3.4.2 with not more than 2 db of degradation. The transmitter shall be capable of producing an equivalent average power during CW or FSK operations.

3.3.3.2 Duty cycle. The AN/PRC-47 shall be capable of operating on the following duty cycles at the maximum specified temperature.

(a) Voice or FSK, high power - Nine minutes receive, 1 minute transmit for an indefinite period.

(b) FSK, high power - Nine minutes receive, 2 minutes transmit for 1 hour.

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(c) FSK, low power - Five minutes receive, 10 minutes transmit for 1 hour.

3.3.3.3 Modulation fidelity. The overall modulation response of the transmitter shall be equal or better than audio response of the receiver. The microphone amplifier shall incorporate peak envelope compression and clipping. With normal microphone input, 0.1 v from a 50.ohm source, the audio signal shall be clipped at least 8 db. Compressor action shall maintain a relatively constant input to the clipper with signal input variations from -6 db to +20 db relative to the normal input.

3.3.3.4 Speech overload. The transmitter shall include the proper circuitry to prevent overload or over-modulation of the RF output.

3.3.3.5 Spurious output.

3.3.3.5.1 Third order intermodulation products. Third order modulation products outside the equipment passband shall be down at least 30 db from either of two equal test tones under standard conditions. At other than standard conditions, the intermodulating distortion may degrade to -20 db from either of two equal test tones; the test signals being adjusted to provide full specified RF power output in the respective modes.

3.3.3.5.2 Undesired sideband. The output level of the lower sideband shall be at least 60 db below that of the upper sideband.

3.3.3.5.3 Carrier suppression. The carrier shall be suppressed at least 40 db below the output of either tone of a two-tone test signal, equal tone amplitudes which shall drive the transmitter to rated output. At the temperature extremes of -40° C and +60° C ambient, the carrier suppression may degrade to -30 db.

3.3.3.5.4 Other spurious output. Second harmonic output shall be at least 35 db below the transmission level and all other spurious radiation outside the passband shall be at least 50 db below the transmission level.

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3.3.4 Antenna coupler.

3.3.4.1 Capability. The antenna coupler and loading device shall be capable of matching the output impedance of the transmitter to the input impedance of the designed whip and counterpoise system or to a 50 ohm load at any frequency within the operating range of the equipment.

3.3.4.2 Efficiency. The antenna coupler efficiency shall be sufficient to provide the following transmitter power versus frequency to a standard 15-foot whip antenna dummy load having the impedance characteristics shown.

Freq, MHz	Load	Minimum power, PEP, w	
		High power	Low power
2.000	4-j1000	20	4
4.000	4-j490	32	6
6.000	6-j285	48	9
8.000	10-j190	60	12
10.000	14-j145	70	14
12.000	20-j100	76	15

3.3.5 Power supply polarity. The dc power supply shall have the negative side grounded.

3.4 Physical details of components.

3.4.1 Transceiver.

3.4.1.1 Form factor. The transceiver dimensions shall not exceed 7-7/16 by 13-1/2 by 21-11/16 inches. High pitch threads shall be used on the screws that attach the legs to the transceiver case.

3.4.1.2 Controls. The following controls shall be provided on the front panel of the transceiver:

(a) Frequency Selection (3 knobs)

(b) Power Amplifier Tune

(c) Power Amplifier Load

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(d) Tune/Operate

(e) High Power/OFF/Low Power

(f) Power OFF/Power ON

Lights OFF

Lights Low

Lights High

(g) Voice/FSK-CW

(h) Volume

(i) Meter Adjust

(j) Battery Test and Thermal Override

Lights shall be provided for the frequency readout dials and for the tuning indicator meter. All controls shall be capable of operation by an operator wearing knit liner and leather shell gloves.

3.4.1.3 Panel markings. Each control shall be marked to aid operating personnel and brief operating instructions shall be located inside the control panel cover.

3.4.2 Battery power supply. The battery power supply (not supplied by the contractor as part of the AN/PRC-47 shall be a 16-cell silver-zinc battery, BB-451/U.

3.4.3 Antenna systems.

3.4.3.1 Whip antenna. The basic antenna system shall consist of a 15-foot whip antenna, AS-1320/PRC-47, and a radial counterpoise system. The whip antenna shall be sectionalized into 24-inch lengths for carrying with the battery power supply. High pitch threads shall be used to connect the sections. The lowest section will be covered with a silicone rubber insulating pad required for operator safety. The counterpoise system shall consist of eight radials attached to the support legs. The support legs shall have provisions for rapidly securing the radials when not in use. The whip antenna shall be insulated and designed to produce the maximum radiated power from the transmitter power output.

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3.4.3.2 Long wire antenna. An antenna, AS-1321/PRC-47, shall be supplied. This antenna consists of an insulated, sectionalized, long wire with jumpers to facilitate changes in length. Sections shall be marked to indicate the number required for various frequencies.

3.4.4 Accessories.

*3.4.4.1 Battery power cable. A power cable assembly, CX-8395/PRC-47, for use with the battery power supply shall be provided. It shall consist of 3 feet of 2-conductor, 14-gage, rubber covered cable terminated to mate with the power input receptacle of the transceiver and a PTO6W-14-5P or equal connector for connection to the adaptor, battery terminal, MX-4430/PRC-47 as indicated on Drawing No. DL 5116-522-2433-011.

3.4.4.2 Vehicular power cable. A power cable assembly, CX-8394/PRC-47 for use with the nominal 26.5 v, db, vehicular power supply shall be provided. It shall consist of 10 feet of 2-conductor, 14-gage, rubber covered cable terminated with a plug to mate with the power input receptacle of the transceiver and terminal clips to attach to the vehicular power supply.

3.4.4.3 AC power cable. A power cable assembly, CX-8393/PRC-47, for use with a single-phase, 400 Hz, ac, power source shall be provided. It shall consist of 50 feet of 2-conductor, 14-gage, rubber covered cable terminated by a plug to mate with the power input receptacle of the transceiver and terminal lugs to attach to the 400 Hz power source.

3.4.4.4 Audio and telegraph accessories. Audio and telegraph accessories shall be provided and shall consist of one each of the following:

(a) Handset, H-33 ( ) /PT

(b) Headset, H-233/PRC-47

(c) Key, telegraph, J-45 (with leg clamp)

(d) Cable assembly, special purpose, electrical, branched, CX-8396/PRC-47. A waterproof audio cable terminated with a plug, U-79/U, and connections to mate with the key and headset.

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(e) Adapter, cable to connector, U-239/PRC-47, terminating in a U-79/U connector and a metal tube containing four conductors with four spring loaded plugs to accept WD-1/TT wire for use with a FSK.

(f) An LS-166/U speaker.

*3.4.4.5 Rucksack and rucksack frames. Two rucksack frames shall be provided with each radio set. An accessories rucksack, antenna case, and electrical equipment legs case shall also be provided. The flaps on the accessory rucksack, antenna case and legs case shall be cuplike and shall completely cover the openings. Rucksack frames shall be furnished in accordance with IP/DES S-42-7

3.4.5 Transit case. The transit case, CY-3700/PRC-47, shall be a watertight case to enclose and protect the receiver-transmitter, R-671/PRC-47, three batteries, BB-451/U, antennas and all accessories during any form of field transportation or storage. It shall be capable of protecting all of the enclosed components under the conditions indicated in 3.2.3.1 and 3.2.3.2. The transit case shall be of rugged construction, lightweight material, compartmentalized and contain shock absorbing material to provide maximum protection to the components enclosed. The transit case shall be equipped with an easily removable cover which shall provide access to all components and a cover retaining bracket. Folding handles shall provide hand holds for carrying.

3.5 Government furnished material (GFM). The Government will furnish two radio sets, AN/PRC-47, for use as models. In order to verify that these models meet the performance requirements of this specification, the contractor shall notify the contracting officer within 75 days after receipt of the GFM of any area of nonconformance. If no notification has been made at the end of this period, the GFM will be presumed to meet all requirements herein.

3.6 Item list. Table II lists the components required for the complete AN/PRC-47 radio set in accordance with Drawing No. 5116-522-2433-011.

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Table II - Components List

Qty per radio set	Item	FSN	Supplier
1	Receiver-transmitter R-671/PRC-47	5820-082-1599	Contr
1	Antenna, AS-1320/ PRC-47 w/case	5985-087-2326	Contr
1	Antenna, AS-1321/ PRC-47	5985-087-2305	Contr
1	Handset, H-33( ) /PT	5965-163-9947	Govt
1	Handset, H-233/ PRC-47	5965-985-3589	Contr
1	Key, telegraph J-45	5805-171-3370	Govt
1	Loudspeaker, LS-166/U	5965-243-6420	Govt
1	Adapter, cable to connector, U-239/ PRC-47	5820-970-6766	Contr
1	Adapter, battery terminal, MX-4430/PRC-47	6135-087-2301	Contr
2 sets	Electrical equip-ment legs, MT-2786/PRC-47, w/case	5820-062-4758	Contr
1	Cable assy, CX-8393/PRC-47	5995-087-2324	Contr
1	Cable assy, CX-8394/PRC-47	5995-082-0487	Contr
1	Cable assy, CX-8395/PRC-47	5995-087-2325	Contr

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Table II - Components List (cont'd)

Qty per radio set	Item	FSN	Supplier
1	Cable assy, CX-8396/PRC-47	5995-087-2327	Contr
2	Rucksack frame	8465-558-0151	Govt
2	Rucksack frame waist strap	8465-024-7281	Govt
1	Rucksack		Contr
1 set	Frame accessories		Contr
1	Case, CY-3700/PRC-47	5820-062-6748	Contr

3.7 Workmanship. The equipment covered by this specification, including all parts and accessories, shall be manufactured and finished in accordance with MIL-E-16400 and the GFM.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection and tests. Inspection and tests shall be conducted at a laboratory designated by the contractor and approved by the contracting officer and shall consist of such tests as required to demonstrate that the first articles submitted meet all the requirements of this specification. First article tests will be witnessed by a Government representative.

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4.2.1 Tests. Table III lists the tests required for first article, production, and sample testing.

4.2.1.1 Inspection and tests. The detailed procedures for conducting tests shall be prepared by the contractor and submitted to the contracting officer for review and approval one month prior to the initiation of any official testing.

4.2.1.2 Tests to be performed. The following quantity of radios shall be tested:

(a) First article tests. Two first article radio sets shall be used to complete these tests. Tests completed by one radio need not be repeated by the other. The Government representative will select which radio to perform each test.

(b) Production tests. All radio sets shall be subjected to the production tests. The requirements of 3.2.4.3 apply to all models; however, to facilitate production testing, the production tests may be conducted in a manner so that a linear extrapolation of measurements taken on a shorter time interval will project the 60 day stability measurements. The requirements of 3.2.4.3 also apply to these projected measurements. The time interval required for these measurements will be determined by the testing procedure used, and will be subject to approval by the Government.

(c) Sample tests. Sample test radio sets will be randomly selected by the Government inspector.

Quantity to be tested	Quantity of radios to be produced
1	1-49
2	50-99
3	100-199
4	200-299
5	300-400

(One additional sample for each additional 100 radios)

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Table III - Tests

Test	Req. Para	Standard cond.	Lo temp(-40° C)	Hi temp	Humidity	Watertight	Vibration	Shock	Pri Power Var	Splashproof	Temp shock	Altitude	Salt spray	Sand & dust	Interference	Lo temp(-25° C)
Sensitivity	3.3.2.1	FA S P	FA S	FA S	FA S		FA S	FA S	FA S		FA		FA	FA		FA
AGC	3.3.2.4	FA S P	FA	FA	FA			FA	FA		FA					FA
Selectivity	3.3.2.2	FA S P	FA	FA	FA			FA			FA					
IF rejection	3.3.2.3.1	FA S P	FA	FA	FA											
Image rejection	3.3.2.3.2	FA S P	FA S	FA S	FA S											
Audio distortion	3.3.2.5.2	FA S P	FA	FA	FA			FA	FA		FA					FA

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Table III - Tests (cont'd)

Test	Req. Para	Standard cond.	Lo temp(-40° C)	Hi temp	Humidity	Watertight	Vibration	Shock	Prim pwr variation	Splashproof	Temp shock	Altitude	Salt spray	Sand & dust	Interference	Lo temp(-25° C)
Cross modulation	3.3.2.3.3	FA														
Volume control	3.3.2.6	FA P														
Power output	3.3.4.2	FA S P	FA S	FA S	FA S		FA S	FA S	FA S		FA	FA	FA	FA		FA
Freq. stability	3.2.4.3	FA S P	FA S	FA S	FA S		FA S	FA S	FA S		FA					FA
Intermod distortion	3.3.3.5.1	FA S P	FA	FA	FA				FA							
Carrier suppr	3.3.3.5.3	FA S P	FA	FA	FA				FA							

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Table III - Tests (cont'd)

Test	Req. Para		Standard cond.	Lo temp(-40° C)	Hi temp	Humidity	Watertight	Vibration	Shock	Prim pwr variation	Splashproof	Temp shock	Altitude	Salt spray	Sand & dust	Interference	Lo temp(-25° C)
Undesired sideband suppr	3.3.3.5.2	FA															
		S	FA	FA	FA	FA				FA							
		P	FA														
Other spurious output	3.3.3.5.4	FA															
		S	FA	FA	FA	FA				FA							
		P	FA														
CW telegraphy operation	3.2.11.2	FA															
		S	FA	FA	FA	FA											
		P	FA														
Side tone	3.2.13	FA															
		S	FA	FA	FA	FA											
		P	FA														
Modulation fidelity	3.3.3.3	FA															
		S	FA	FA	FA	FA											
		P	FA														
Duty cycle	3.3.3.2				FA												

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Table III - Tests (cont'd)

Test	Req Para	Standard cond.	Lo temp(-40° C)	High temp	Humidity	Watertight	Vibration	Shock	Prim pwr variation	Splashproof	Temp shock	Altitude	Salt spray	Sand & dust	Interference	Lo temp(-25° C)
Mechanical & visual insp.	Various	FA S P	FA S	FA S	FA S	FA	FA S	FA S		FA S	FA	FA	FA	FA		
Power input	3.2.10	FA S P														
Talk out	Various	P								FA S						
Battery operation	3.3															FA
High pot(cables)	3.4.4.1	FA S P	FA	FA	FA											
Interchangeability		FA														
Channelization	3.3.1	FA														

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Table III Tests - (cont'd

Test	Req Para	Standard cond.	Lo temp(-40° C)	High temp	Humidity	Watertight	Vibration	Shock	Prim pwr variation	Splashproof	Temp shock	Altitude	Salt spray	Sand & dust	Interference	Lo temp(-25° C)
Audio power output	3.3.2.5.1	FA S P														
Audio response	3.3.2.5.3	FA														
Remote control	3.2.15	FA														
Item to be tested RT-671/PRC-47		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
CY-3700/PRC-47		X				X	X	X								
Antennas		X	X	X	X		X	X								
Power cables		X	X	X	X		X	X								
Accessories		X					X	X								

Note: FA - First article tests
S - Sample tests
P - Production tests

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4.2.2 Shock and vibration tests.

4.2.2.1 Vibration.

4.2.2.1.1 Equipment nonoperating. The RT-671/PRC-47 shall have recorded measurements at standard conditions prior to vibration. A complete AN/PRC-47 radio set shall be enclosed in the transit case and subjected to the following tests:

(a) The transit case shall be rigidly mounted on the vibration machine and shall be subjected to vibration in the frequency range of 10 to 55 Hz at 5.0 gravity (g) or 0.060-inch total excursion, whichever is the limiting value.

(b) Frequency change shall be at the rate of less than 1.0 Hz.

(c) Duration of the test shall be one-half hour in each of three planes.

(d) There shall be no rigid body resonance below 55 Hz.

(e) At the end of the vibration test in all three planes, the RT-671/PRC-47 shall be removed from the transit case and connected to a 24-v, dc, source. Measurements of power output at high and low operation, frequency accuracy at 2 frequencies, and receiver sensitivity shall be recorded. These readings shall show no deterioration from readings obtained prior to the vibration test.

4.2.2.1.2 Equipment operating. The receiver-transmitter, RT-671/PRC-47, shall be rigidly mounted on the vibration machine and have power applied from a 24-v, dc, source. The operating unit shall be subjected to vibration in the frequency range of 10 to 55 Hz at 5 g's or 0.060-inch total excursion, whichever is the limiting value. Frequency change shall be at a rate of less than 1.0 Hz. Duration of the test shall be one-half hour in each of three planes. There shall be no body resonance below 55 Hz. Readings of power output, sensitivity, and frequency stability shall be monitored throughout the test in all three planes. One reading of each of the listed performance parameters shall be recorded in each of the vibration planes at the vibration frequency ranges indicated below.

(a) 10 to 20 Hz
(b) 20 to 30 Hz
(c) 30 to 40 Hz
(d) 40 to 55 Hz

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Upon completion of the vibration test in all three planes, the indicated performance parameters shall be measured and recorded.

4.2.2.2 Shock test. One model of each, the RT-671/PRC-47; battery power supply, BB-451/U; and a transit case with a complete system enclosed, shall withstand a 3-hour test on the LAB package testing machine. For this test, the unit shall be mounted rigidly to an adapter plate, secured to the machine. The machine shafts shall be operated in phase and adjusted in speed to produce average peak accelerations of 10 g's, with occasional peaks of 20 g's as measured on the case of the item. At the completion of 3-hours shock testing, remove the equipment from the transit case, and apply power from a 24-v dc, source. Record the performance parameters required under shock test.

4.3 Splashproof test. Place equipment in operating condition and test in accordance with Table I of MIL-STD-108 for splashproof equipment. Water accumulation is limited to 1 tablespoonful inside the case providing the radio set is operable prior to removing the water.

4.4 Watertight test. Place equipment in the transit case and test in accordance with Table III of MIL-STD-108 for watertight equipment. Repeat same test on transceiver with transceiver case cover in place.

4.5 High temperature test. The equipment shall be tested in accordance with Procedure II, of MIL-E-5272 with the following exceptions:

(a) Test operation of equipment at +60° C using specified battery.

(b) Maximum temperature to be +65° C.

4.6 Low temperature tests. The equipment shall be tested in accordance with Procedure II, of MIL-E-5272 with the following exception:

(a) After 72 hours at -62° C the temperature will be raised to -40° C to test operation using a 24-v external source. The temperature will then be raised to -25° C and the equipment will be tested utilizing a 16-cell silver-zinc battery that has been soaked for 12 hours at -25° C.

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*4.7 Temperature shock test. The equipment shall be tested in accordance with Procedure I, of MIL-E-5272 except that the high temperature shall be +65° C.

4.8 Humidity test. The equipment shall be tested in accordance with Procedure I, of MIL-E-5272 with the following exceptions:

(a) Maximum temperature shall be +65° C.

(b) The test will be concluded at the end of 3.0 cycles.

(c) Moisture may be removed by wiping.

4.9 Altitude test. The equipment shall be tested in accordance with Procedure VI, of MIL-W-5272 to 12,000 feet at -40° C.

4.10 Salt spray test. Place equipment in test chamber in operating condition and perform the test in accordance with MIL-E-5272.

4.11 Sand and dust. Place equipment in operating condition in test chamber and perform the tests in accordance with Procedure III, of MIL-E-5272.

*4.12 Interference test. The equipment shall be tested for conducted interference, radiated interference, and receiver front end rejection in accordance with class 1 of MIL-STD-461. Testing shall be performed in accordance with the applicable test methods of MIL-STD-462.

4.13 Test reports. The supplier shall prepare and submit to the procuring activity copies of all test reports in duplicate not later than 30 days after completion of tests.

4.14 Approval. Approval of the two first article samples shall be by the contracting officer upon satisfactory completion of all the tests.

4.15 Acceptance. With the exception of the first two samples, acceptance of all items shall be made by the Government inspector.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Preservation, packaging and packing shall be level A or C as specified in the contract or order in accordance with MIL-E-17555 (see 6.2).

5.2 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. Radio sets are employed as a team pack high frequency single sideband communication system used primarily by tactical units.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Whether first article is required (see 3.1).

(c) Level of preservation, packaging, and packing (see 5.1).

6.3 Color. Color chips for Marine Corps green color No. 23 may be obtained from the Commandant of the Marine Corps (Code CSY-10), Headquarters U. S. Marine Corps, Washington, D. C. 20380.

*6.4 First article. The first article shall be examined and tested for approval, at the contractor's plant or at an independent commercial testing laboratory acceptable to the procuring activity. First article tests shall be witnessed by a Government representative of the procuring activity. Approval of the first article shall be by the contracting officer.

6.5 Marginal notations. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are

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Review activity:
Air Force-71
User activities:
Army-EL
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Preparing activity:
Navy - MC
Project No. 5820-0046

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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